



Project Title: 📊 Customer Sales Analysis Dashboard

Duration: 2–3 Hours

🎯 Objective

You are provided with a dataset of 150 customer purchases. Using your knowledge of NumPy, Pandas, Matplotlib, and Seaborn, perform data cleaning, analysis, and visualization to understand customer behavior, sales trends, and product popularity.

📁 Dataset:

`customer_sales_data.csv` (generate from AI tool)

Fields include:

- Customer_ID
 - Gender
 - Age
 - Region
 - Product_Category
 - Units_Purchased
 - Price_Per_Unit
 - Purchase_Date
 - Total_Amount
-



Tasks Checklist:



Data Preparation

1. Load the dataset and display first 5 rows.
2. Show column names, data types, and shape of the dataset.
3. Convert Purchase_Date to datetime.
4. Use NumPy to:
 - Calculate average Total_Amount and Units_Purchased.
 - Find the highest and lowest purchase amount.



Data Analysis

5. Group total sales by:
 - Gender
 - Region
 - Product Category
6. Show month-wise or weekly total purchases using Purchase_Date.



Visualization

7. Plot a bar chart of total sales by Product Category.
8. Plot a pie chart of customer distribution by Region.
9. Plot a line chart of daily total sales over time.
10. Use Seaborn:
 - Create a box plot of Total_Amount by Gender.
 - Create a heatmap of correlation between numeric columns.



Submission

- Python script or Jupyter notebook with clean code, comments, and all tasks completed.
- Each graph should have appropriate titles, axis labels, and styling.